

Unsupervised Learning of Visual Features by Contrasting Cluster Assignments

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A. Summary

This paper proposes SwAV (Swapping Assignments between Views), an online clustering-based self-supervised method for visual representation learning. Instead of comparing features of different views directly as in contrastive learning, it assigns each view to learnable prototypes and predicts the cluster assignment of one view from the feature of another view of the same image. Together with multi-crop, a data augmentation strategy that adds low-resolution crops at little extra cost, SwAV achieves 75.3% top-1 accuracy on ImageNet linear evaluation with ResNet-50 and surpasses supervised pretraining on all considered transfer tasks.

B. Strengths

1. Memory efficient and flexible in batch size. SwAV avoids pairwise feature comparisons and requires neither a large memory bank nor a momentum encoder. It stores only about 3.8K features and works well with both large (4096) and small (256) batches, making it usable on GPUs with small VRAM.
2. Effective multi-crop augmentation. Adding several low-resolution crops to the two standard views increases the number of views at little extra memory cost, and consistently improves SimCLR, DeepCluster, and SeLa by 2-4% top-1 accuracy.
3. Truly online method. Unlike DeepCluster and SeLa, SwAV computes codes within each mini-batch. This allows it to scale to unlimited data, as shown by pretraining on 1B uncured Instagram images.
4. Comprehensive experiments. SwAV is evaluated on ImageNet and multiple downstream tasks, surpassing supervised pretraining on all of them, with a fair ablation against contrastive methods.

C. Major Weaknesses

1. Detail the important weaknesses you identified and justify them. For example:
2. If you found the method difficult to understand, explain what the authors did wrong and give examples.
3. If the claimed contributions are not backed up by the experiments, explain what should have been done.

4. Be specific: Do not simply give summary judgments. Writing “not novel”, “unclear”, “incorrect” is not sufficient!
5. Your comments will be much more helpful to the ACs and the authors than your score.
6. (2026) Suppose this paper was submitted in 2026, what would be its major weaknesses from today’s point of view?
7. (2026) In other words, this comment is based on current technological progress, not the time when the paper was originally published.

D. Minor Weaknesses

1. Mention the weaknesses that are easy to fix in a revision of the paper. For example:
2. Missing references that should be discussed but do not really change the novelty of the paper.
3. Occasional typographic errors.
4. Minor unclear points.
5. Mention here any minor suggestions, questions, corrections, etc. that can help the authors improve their paper.
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E. Justification

Balance the strengths and the weaknesses to explain why the paper should be accepted or not. Does the paper present sufficient knowledge advancement? Use your best judgment.

(2026) Suppose this paper was submitted in 2026, on what grounds would you recommend accepting or rejecting it from today’s point of view? In other words, this comment is based on current technological progress, not the time when the paper was originally published.

F. Final Rating

5: Strong Accept

4: Weak Accept

3: Borderline

2: Weak Reject

1: Strong Reject

Provide (2026) final rating as well.

This template follows the review format of CVPR
2025 [1] and [2].

G. Reflections

Share your reflections on this paper. What did you
learn (technologies, writing, etc.)? How did it inspire
you? Is it useful for your project or research? How
might you apply it to your research? Just examples.
Feel free to connect the paper to yourself in other ways.

References

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